Capacity To Ship Packages Within D Days (medium)

Problem Statement

You are given `weights` array of size `n`, where `weights[i]` represents the weight of the ith package. These `n` packages should be shipped from one place to another place via one ship within a given `days`.

If you can load the ship with some packages each day (in the order given in `weights[i]`), find the smallest possible maximum weight the ship can carry per day without exceeding the given number of days for shipping all packages.

Examples

Example 1:

- **Input**: weights = `[10,20,30,40,50]`, days = `3`
- **Expected Output**: `60`
- **Justification**: A capacity of 60 means shipping [10,20,30] on day 1, [40] on day 2, and [50] on day 3.

Example 2:

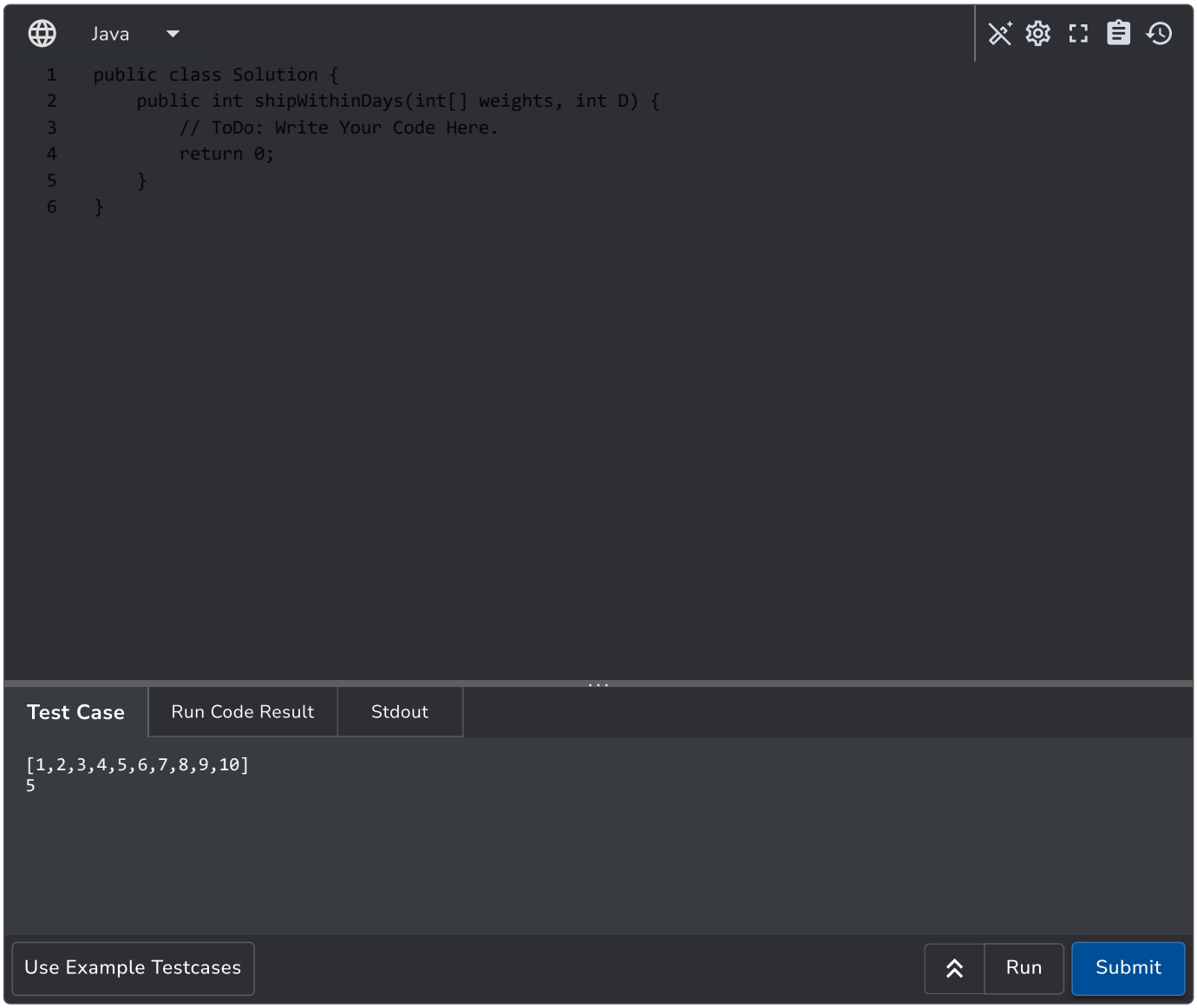
- **Input**: weights = `[5,5,5,5,5,5,5,5,5,5]`, days = `5`
- **Expected Output**: `10`
- **Justification**: A capacity of 10 allows shipping two packages of weight 5 each day, perfectly fitting within 5 days.

Example 3:

- **Input**: weights = `[4, 8, 5, 2, 6, 3, 7, 9]`, days = `4`
- **Expected Output**: `13`
- **Justification**: A capacity of 13 allows shipping [4, 8] on day 1 (total 12, close to 13 without exceeding), [5, 2, 6] on day 2 (total 13), [3, 7] on day 3 (total 10), and [9] on day 4.

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Solution: Capacity To Ship Packages Within D Days

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